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CONSEJO DE CIENCIA Y TECNOLOGÍA
DEL ESTADO DE NAYARIT

LAB 1 Logical Comparators

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Technical Report

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Abstract: *FPGA Terasic*

Board – ALTERA Cyclone IV EP4CE115F29C7N

IDE – Quartus Prime 18.1

Boolean Algebra

Digital Gates & Logical Comparators

Git & GitHub Control Version

Keywords: *Logical Comparators, Schematic, Bit, Input, Output, Switches & Leds.*

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1.- INTRODUCTION

This practice involved developing a logical comparator project using **Quartus Prime 18.1.0** and the **DE2-115** board with its **Cyclone IV EP4CE115F29C7 FPGA**. After reviewing key concepts like Boolean algebra, logic gates, and digital systems, the software work began. Emphasis was placed on creating custom schematic blocks for program generation and testing. The project was carried out in stages: designing logic with gates, assigning nets and labels, configuring switches and LEDs, and finally uploading the program via **USB Blaster** to verify its functionality.

1.1.- Requirements

Design and implement a schematic-based circuit that takes two input values (A and B) and determines whether:

- $A < B$ -- LEDG[2]
- $A == B$ -- LEDG[1]
- $A > B$ -- LEDG[0]

The implemented circuit must have the following characteristics:

- Inputs "A" and "B" must be 4 bit values each one.
- Outputs must be 3 bits values that indicate which condition ($A < B$, $A > B$, $A == B$).
- PIN Assignment must be for "A" SW[17] to SW[14].
- PIN Assignment must be for "B" SW[0] to SW[3].

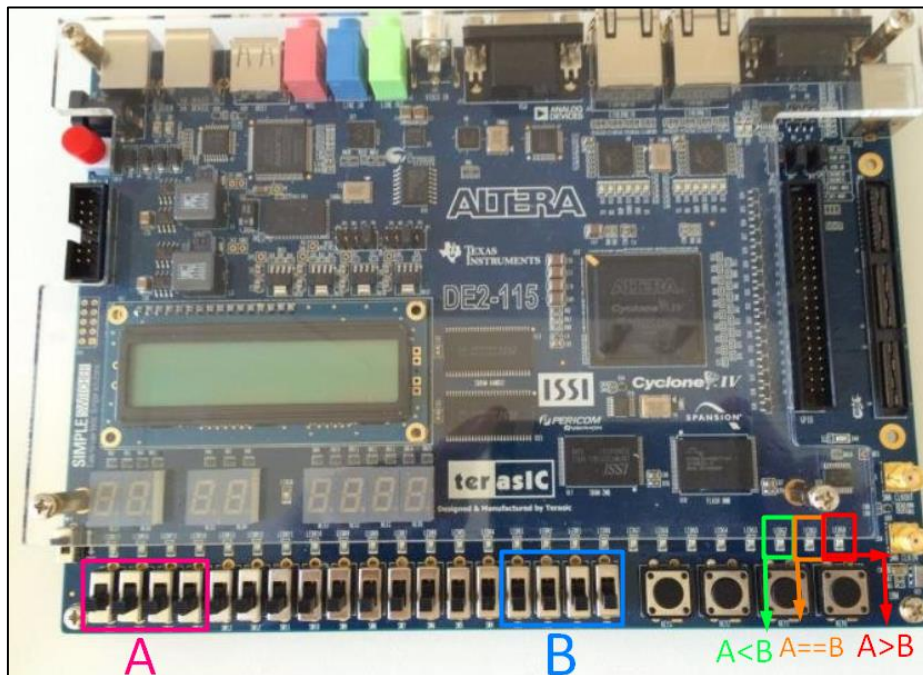


Figure: 1 PinAssignment

2.- FUNCTIONAL DESCRIPTION

The project was created with the help of the wizard, and the corresponding FPGA was then selected.

Family, Device & Board Settings

Device | Board

Select the family and device you want to target for compilation.
You can install additional device support with the Install Devices command on the Tools menu.

To determine the version of the Quartus Prime software in which your target device is supported, refer to the [Device Support List](#) webpage.

Device family
Family: **Cyclone IV E**
Device: All

Target device
☐ Auto device selected by the Fitter
☒ Specific device selected in 'Available devices' list
☐ Other: n/a

Show in 'Available devices' list
 Package: Any
 Pin count: Any
 Core speed grade: Any
 Name filter: **115**
☒ Show advanced devices

Available devices:

Name	Core Voltage	LEs	Total I/Os	GPIOs	Memory Bits	Embedded multiplier 9-bit elements	PL
EP4CE115F23C7	1.2V	114480	281	281	3981312	532	4
EP4CE115F23C8	1.2V	114480	281	281	3981312	532	4
EP4CE115F23C8L	1.0V	114480	281	281	3981312	532	4
EP4CE115F23C9L	1.0V	114480	281	281	3981312	532	4
EP4CE115F23I7	1.2V	114480	281	281	3981312	532	4
EP4CE115F23I8L	1.0V	114480	281	281	3981312	532	4
EP4CE115F29C7	1.2V	114480	529	529	3981312	532	4

Figure: 2 FPGA_Selection

The logic gates were placed using the libraries preloaded in the IDE.

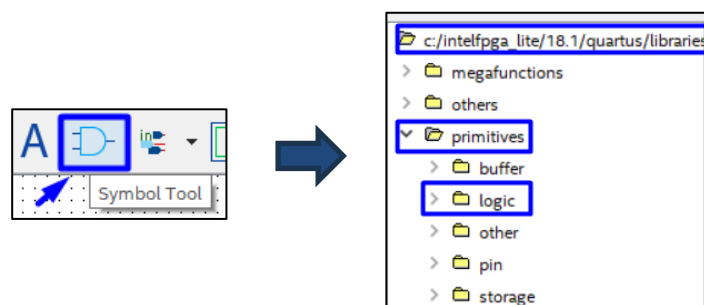


Figure: 3 Logic_Gates

This section was selected for the assignment of inputs and outputs. And the reference for sintaxis A [3..0] for define 4-bit values.

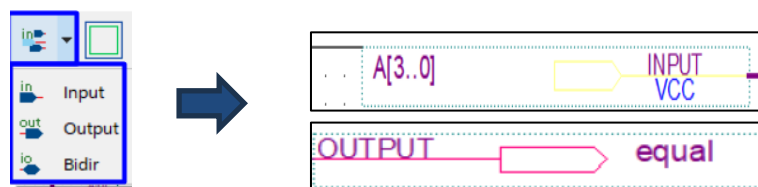


Figure: 4 Inputs Outputs

The type of wiring connection to gates and buses was configured.

1.- Standard Wire.

2.- Bus Wire.



Figure: 5 Wire & Bus Connection.

For this lab, the first step was to define the logic circuit for equality conditions, followed by the logic circuit for the greater-than condition, and finally the one for the less-than condition.

EQUAL STATE ($A == B$)

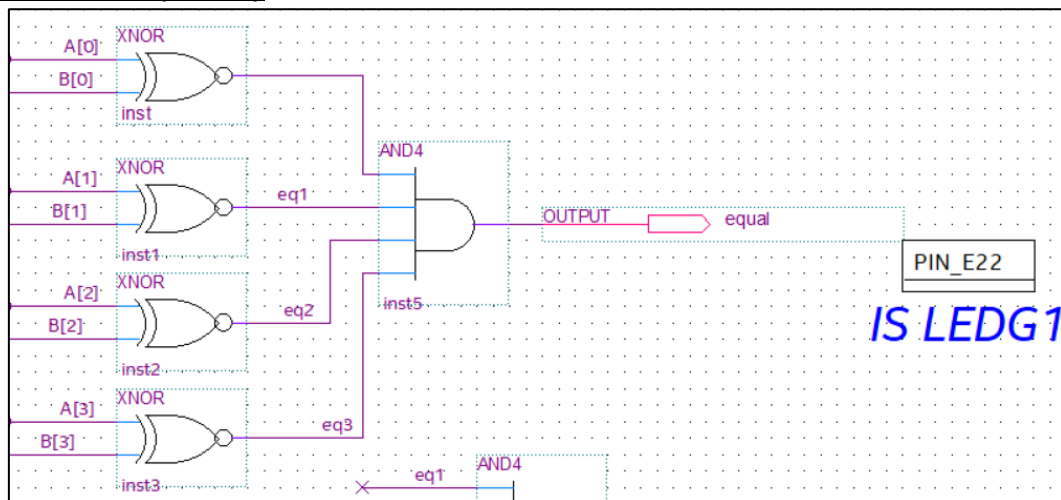


Figure: 6 Equal State

GREATER STATE ($A > B$)

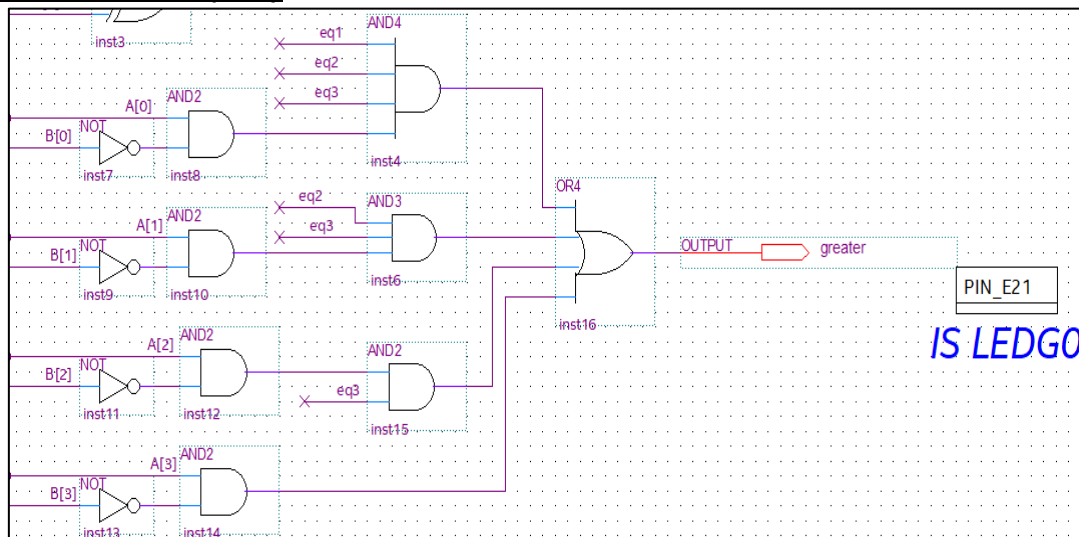


Figure: 7 Greater State

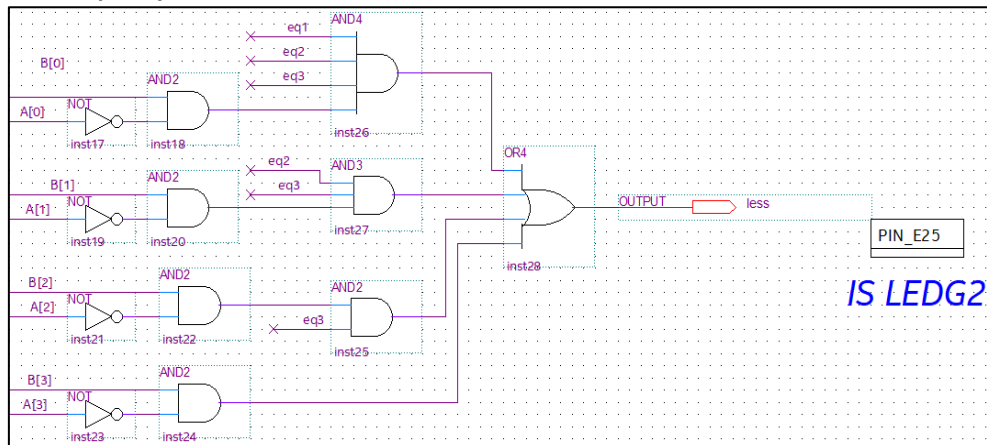
LESS STATE (A<B)

Figure: 8 Less State

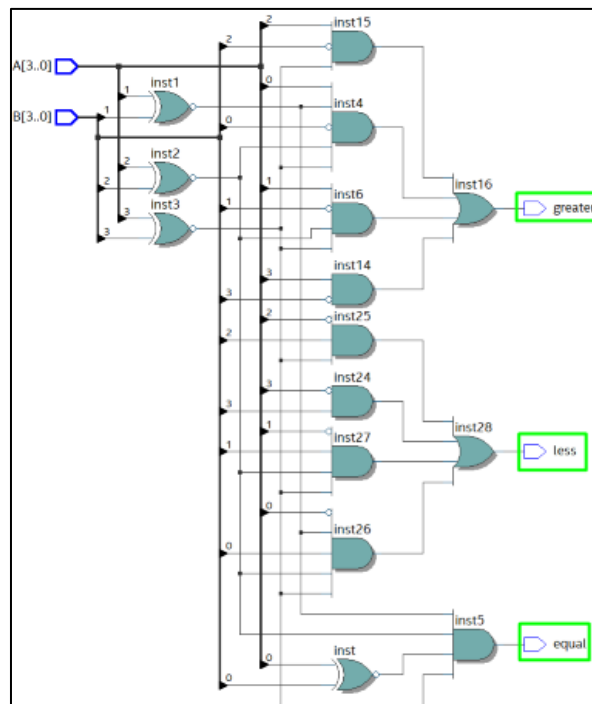
RTL VIEWER

Figure: 9 RTL_Viewver

In “*PinPlanner*” we did the pin assignment for Inputs and Outputs, in the “Location”.

Node Name	Direction	Location
in A[3]	Input	PIN_Y23
in A[2]	Input	PIN_Y24
in A[1]	Input	PIN_AA22
in A[0]	Input	PIN_AA23
in B[3]	Input	PIN_AD27
in B[2]	Input	PIN_AC27
in B[1]	Input	PIN_AC28
in B[0]	Input	PIN_AB28
out equal	Output	PIN_E22
out greater	Output	PIN_E21
out less	Output	PIN_E25

Figure: 10 PinPlanner

Once the logic circuit was completed, the program was compiled for verification, with the expectation that no errors would appear in the message terminal.

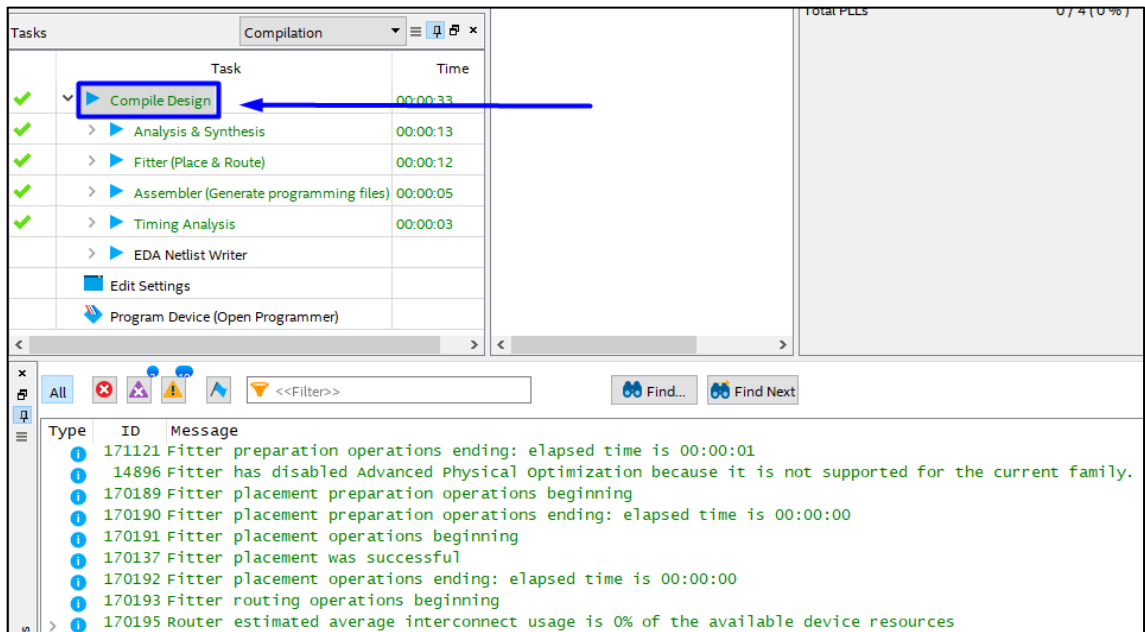


Figure: 11 Compilation

Next, the program is loaded onto the FPGA using the USB-Blaster under the hardware setup configuration.

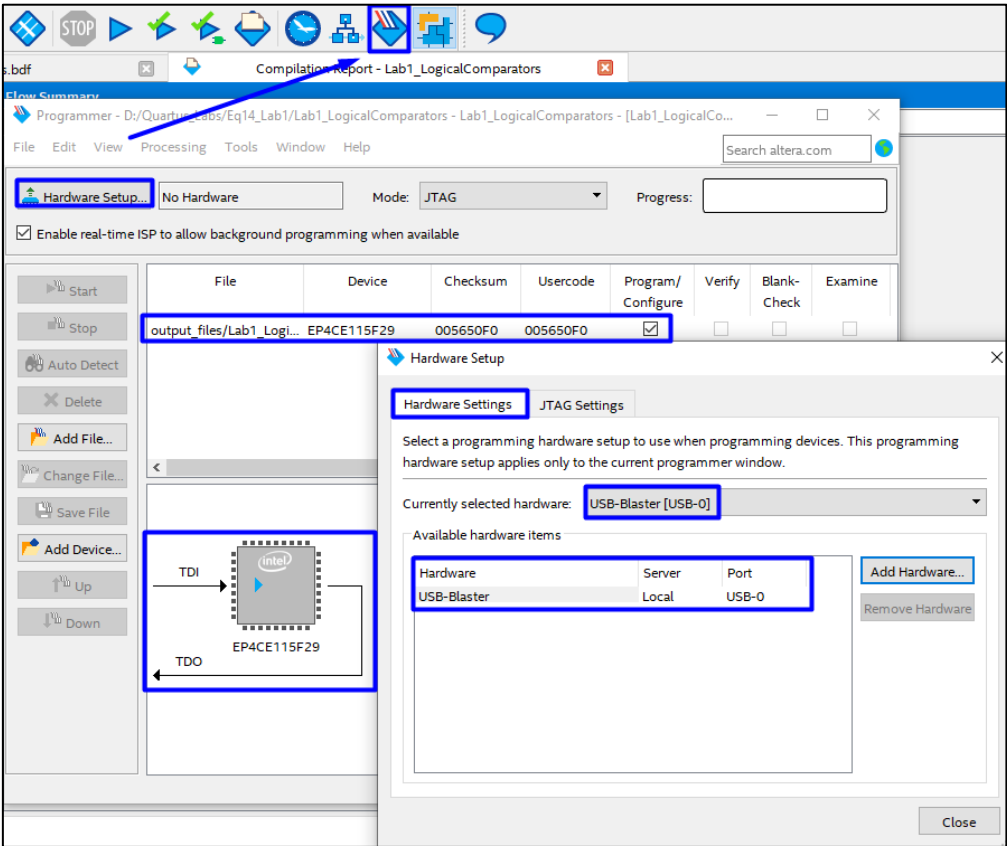


Figure: 12 Programmer/Debug

Then programmer the process to FPGA.

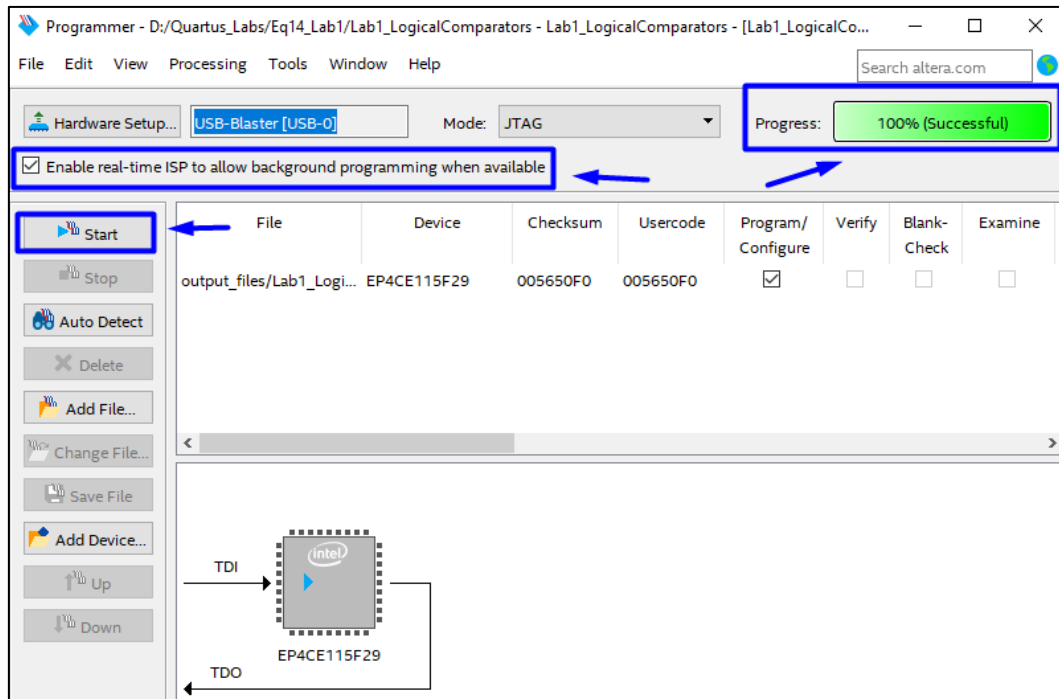


Figure: 13 ProgrammerVerify

2.1.- Boolean Algebra Equations

4-Bit Comparator:

$$A = A_3 A_2 A_1 A_0$$

$$B = B_3 B_2 B_1 B_0$$

$A_3 B_3$	$A_2 B_2$	$A_1 B_1$	$A_0 B_0$	$A < B$	$A = B$	$A > B$
$A_3 > B_3$	x	x	x	0	0	1
$A_3 < B_3$	x	x	x	1	0	0
$A_3 = B_3$	$A_2 > B_2$	x	x	0	0	1
$A_3 = B_3$	$A_2 < B_2$	x	x	1	0	0
$A_3 = B_3$	$A_2 = B_2$	$A_1 > B_1$	x	0	0	1
$A_3 = B_3$	$A_2 = B_2$	$A_1 < B_1$	x	1	0	0
$A_3 = B_3$	$A_2 = B_2$	$A_1 = B_1$	$A_0 > B_0$	0	0	1
$A_3 = B_3$	$A_2 = B_2$	$A_1 = B_1$	$A_0 < B_0$	1	0	0
$A_3 = B_3$	$A_2 = B_2$	$A_1 = B_1$	$A_0 = B_0$	0	1	0

$$A < B = \underbrace{A_3' B_3}_{\text{from 1 bit comp.}} + \underbrace{(A_2 \odot B_2) (A_2' B_2)}_{\text{1 bit comp.}} + (A_3 \odot B_3) (A_2 \odot B_2) (A_1' B_1) + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0' B_0)$$

$$A = B = (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0)$$

$$A > B = A_3 B_3' + (A_3 \odot B_3) (A_2 B_2') + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 B_1') + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 B_0')$$

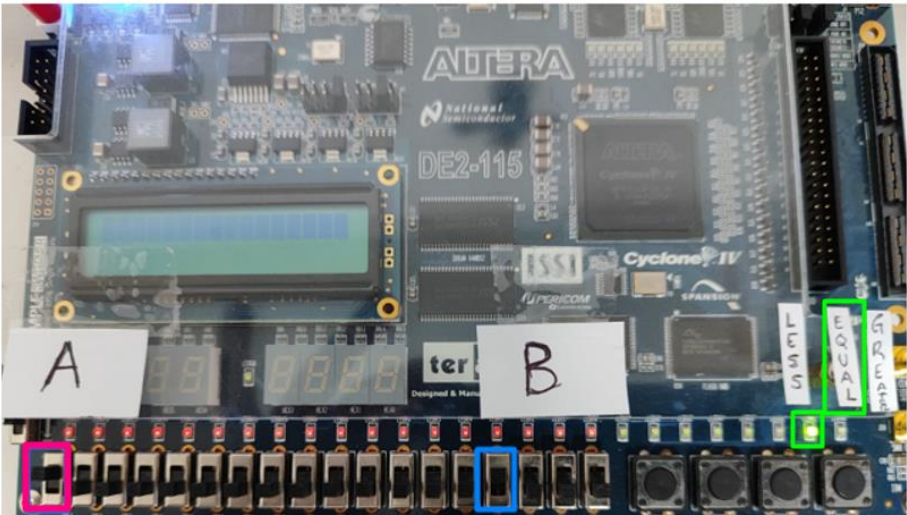
Figure: 14 Equations

2.2.- Results

For this case A [3..0] represents a 4 bits values and B [3..0] also. In this case, "A" represents 4-bit values, and so does "B", resulting in a total of 16 possible values for each. **This leads to 16 x 16 = 256 possible combinations in the truth table.** This is possible taking the LSB and MSB and generating the comparison between every bit for each input.

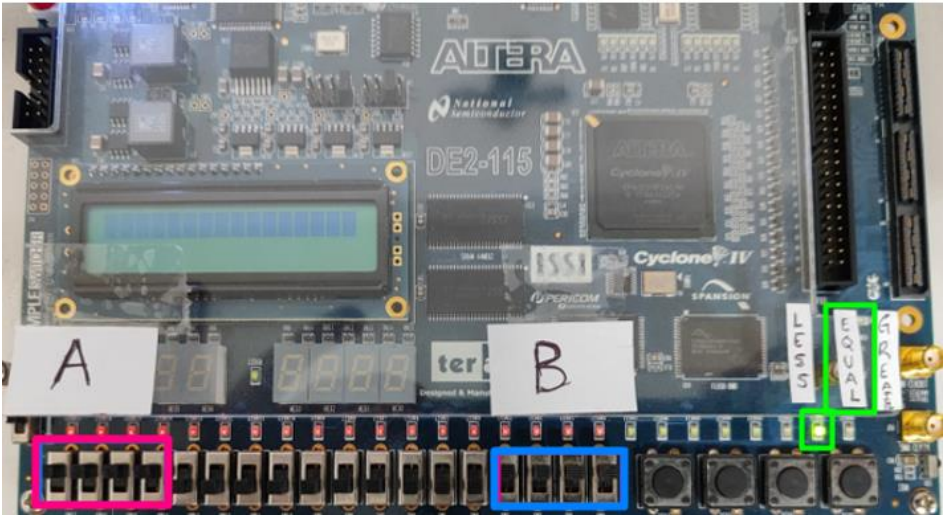
A				B				R
0	0	0	0	0	0	0	0	Equal = Led1
0	0	0	1	0	0	0	1	Equal = Led1
0	0	1	0	0	0	1	0	Equal = Led1
0	0	1	1	0	0	1	1	Equal = Led1
1	1	1	1	1	1	1	1	Equal = Led1

Figure: 15 TrueTable_Equal



A = 1000
B = 1000
LED1 = EQUAL STATE

Figure: 16 Test1_Equal



A = 1111
B = 1111
LED1 = EQUAL STATE

Figure: 17 Test2_Equal

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A				B				R
0	0	0	1	0	0	0	0	Greater = Led0
0	0	1	1	0	0	0	1	Greater = Led0
0	1	1	0	0	1	0	1	Greater = Led0
1	0	0	1	1	0	0	0	Greater = Led0
1	1	1	1	1	1	1	0	Greater = Led0

Figure: 18 TrueTable_Greater

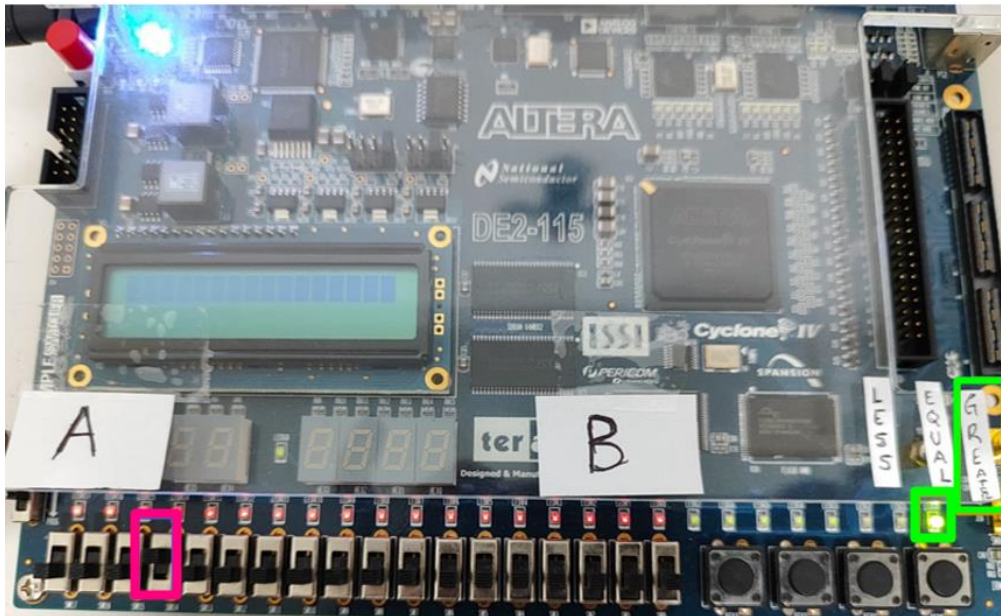


Figure: 19 Test1_Greater

A = 0001
B = 0000
LEDO = GREATER

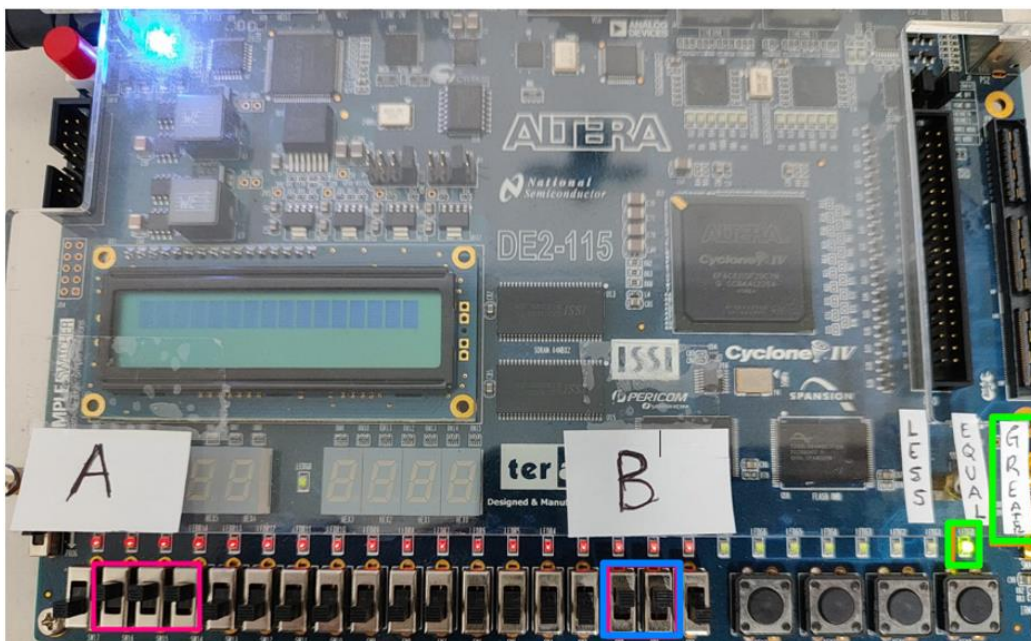


Figure: 20 Test2_Greater

A = 0111
B = 0110
LEDO = GREATER

A				B				R
0	0	0	0	0	0	0	1	Less = Led2
0	0	1	0	0	0	1	1	Less = Led2
0	1	0	0	0	1	1	0	Less = Led2
1	0	0	1	1	0	1	1	Less = Led2
1	1	1	0	1	1	1	1	Less = Led2

Figure: 21 TrueTable_Less



A = 1001
B = 1011
LED2 = LESS

Figure: 22 Test1_Less



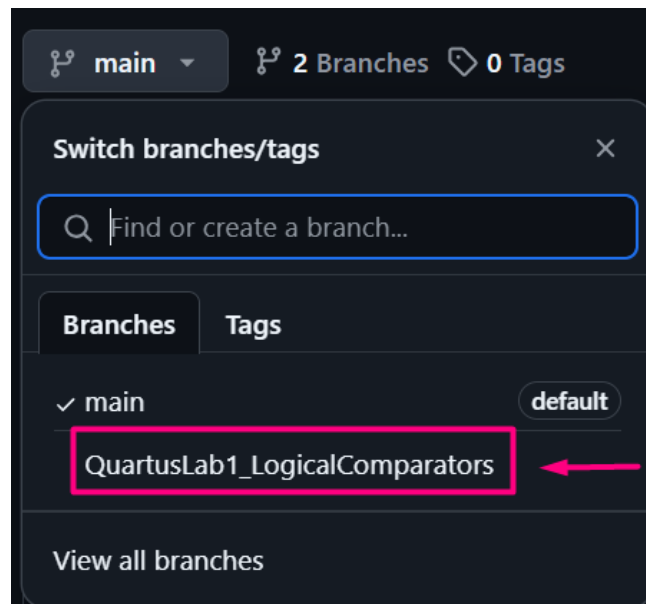
A = 1110
B = 1111
LED2 = LESS

Figure: 23 Test2_Less

2.3.- GitHub Repository

A branch was created in the course repository corresponding to Lab1_LogicalComparators.

<https://github.com/CeicGitHub/Fundamentals Of Digital Design FPGA>



3.- CHALLENGUES/ISSUES

Alonso Emmanuel López Macías

One of the main challenges of the practice was understanding the circuit logic to determine its behavior, including the reasoning and interpretation of Boolean equations. During debugging, knowing how to manually load the driver was extremely helpful. Additionally, learning about how repositories work made remote collaboration much easier.

Cesar Eduardo Inda Cenicerros

In my case, one of the most challenging aspects was revisiting digital logic and relearning key concepts by going back to the study of digital electronics. It also involved understanding how to use Quartus, including its syntax and IDE configurations, defining pin assignments, and grasping the overall logic functionality in the practice. The instructor's lessons were a great help in understanding each topic in greater detail.

4.- CONCLUSIONS

Alonso Emmanuel López Macías

Personally, this practice and course are allowing me to see and analyze the importance of understanding digital systems in greater depth—from the underlying mathematics and algebra to the implementation of functionalities in FPGA devices. It highlights the significance of having a clear understanding of digital design, especially the use of logic gates, as well as many other abstract and implicit concepts involved in each of these implementations.

Cesar Eduardo Inda Cenicerros

Through the practice and classes, I came to realize the importance of having a deep understanding of digital electronics. In my academic career, I had previously studied it in a very general way, without fully grasping how crucial it is to much of the technology we see today. Personally, I'm an embedded systems programmer, and having the opportunity to learn more about these systems has allowed me to understand what lies beneath them in terms of digital design using logic gates. It has also given me greater clarity and understanding of many functions and applications related to my current work. This experience opened my eyes to how broad the use of FPGAs can be.